

SSD_TinyAlsa usage reference

1 Overview

At the beginning of the design of the Audio architecture of SSD20X itself, the Alsa usage scenario was not considered, so currently only TinyAlsa related playback/recording operations are performed. Currently, it does not support complete Alsa operations, such as Dmix operations, etc.

2. Platform alsa support needs to depend on config

1. Alsa needs to rely on the configuration of the kernel. The configuration that the kernel needs to open is as follows (use make menuconfig to configure)

`CONFIG_PM = y`

`CONFIG_SOUND = m`

`CONFIG_SND = m`

`CONFIG_SND_PCM_OSS = m`

`CONFIG_SND_PCM_TIMER = y`

The kernel will generate:

`insmod soundcore.ko`

`insmod snd.ko`

insmod snd-timer.ko

insmod snd-pcm.ko

2. insmod mi_alsa.ko

3. alsa test method: use standard Tinyalsa test

The platform alsa can directly use the standard tinyalsa bin to test, but, because tinyalsa itself is also following the kernel version, for the ALSA adaptation of our SSD platform, it is recommended to use the tinyalsa1.1.1 version test, the following is the tested version of tinyalsa information:

1. Use Tinyalsa's source code to obtain

网址：<https://github.com/tinyalsa/tinyalsa/tree/1.1.1>

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下载：`git clone` <https://github.com/tinyalsa/tinyalsa.git>

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2. tinyalsa编译

配置编译链：

```
export ARCH = arm

export CROSS_COMPILE = arm-linux-gnueabi-

export CC = $(CROSS_COMPILE)gcc
```

直接在tinyalsa根目录执行：

```
make clean;make
```

3. 生成的测试bin

生成目录：tinyalsa/utils bin：tinycap/tinymix/tinypcminfo/tinyplay

4. 测试bin基本测试命令

tinyplay播放测试：

```
./tinypplay *.wav
```

tinypcminfo 查看 pcm 通道的相关信息:

```
./tinypcminfo -D 0
```

tinycap 录音测试 :

```
./tinycap cap_out.wav
```

tinymix 设置和获取音量 :

```
./tinymix contents      //查看对应Dev的相关属性  
./tintmix set 1 60      //设置对应Dev的音量，这里是获取dev=1  
./tinymix get 1         //获取对应Dev的当前音量，这里是获取dev=1
```

4. 平台特性说明

AO说明 :

a. Tinyalsa部分版本限制gain不能设负值，故将AO gain [-60, 30] mapping 至 [0, 90]，如：-60→0，30→90。

b. 若使用mi_alsa、Lineout不支持采样率12k、24k(alsa本身限制)；所支持采样率：8k、11.025k、16k、22.05k、32k、44.1k、48k；Lineout支持1chn单声道或立体声。

c. Alsa/Tinyalsa上层暂时无法下I2S参数：eWorkmode、bSyncClock、eFmt、eMclk，使用I2S时会受限制。